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Data-driven fault detection and diagnosis in industrial process systems: A systematic review and perspective

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A B S T R A C T

Industrial process systems, given their complexity and high risk, can cause catastrophic accidents in the case of failure, leading to casualties, environmental pollution, and economic losses. In the era of Industry 4.0, the rapid progress of intelligent production turns data-driven fault detection and diagnosis (FDD) into a crucial means to ensure the safety and reliability of representative systems, particularly across chemical, petrochemical, and energy industries where benchmark platforms such as the Tennessee Eastman Process (TEP), continuous stirred-tank reactor (CSTR), and Cranfield multiphase-flow facility exemplify practical applications. This paper presents the first systematic review of data-driven FDD research in industrial processes using the Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA) methodology, which focuses on the core objectives of process fault diagnosis, prognosis, predictive maintenance, and condition monitoring. With a structured approach to identify, appraise, and analyze screening literature, this paper thoroughly comprehends the complete pipelines from data benchmarking to modeling and diagnosis. The application of large language model (LLM), multi-source digital twin (DT), and explainable artificial intelligence (XAI) techniques in complex fault pattern recognition is further emphasized, providing practical guidance for the enhancement of system safety and sustainability in the context of smart industry.

1. Introduction

Over forty years after Bhopal concretized the concept of process safety management, incidents such as unplanned releases of toxic substances and even reactor explosions [1,2] continue to impose a heavy burden on the process industries. According to the latest report from the U.S. Chemical Safety and Hazard Investigation Board (CSB), over 500 catastrophic accidents were recorded under federal reporting rules between April 2020 and December 2024. Among the 51 incidents disclosed [3,4], 11 involved at least one fatality, 29 resulted in serious injuries, and the total property damage exceeded $1.7 billion. Marsh Specialty's 28th edition of the loss survey reveals that insured property losses in the hydrocarbon industry alone have exceeded $45 billion over the past five decades [5], a figure that does not include the subsequent effects of plant shutdowns and environmental remediation. These figures demonstrate that, despite regulatory advances and sophisticated automation, the process industry sector continues to operate on a risk plateau.

Through a detailed investigation of loss events statistics, fault detection and diagnosis (FDD) gaps that contribute to the persistence of plateau phenomena have been identified, as shown in Tables 1 and 2. Table 1 presents five representative industrial accidents that defined the formative stage of diagnostic research and process safety regulation, whereas Table 2 highlights five recent loss events illustrating persistent fault detection challenges in modern data-driven operations. These cases are selected against three filters: (i) severity ($\geq$ $100 million insured damage or $\geq$ 5 fatalities); (ii) diagnostic relevance (instrumentation, control systems, or maintenance errors implicated in the initiating event); (iii) recorded root cause available in databases maintained by the CSB or Marsh Specialty. Historical incidents reveal that the lack of real-time fault detection or predictive maintenance oversight can lead to the unchecked spread of potential deviations [6], such as the Flixborough temporary bypass and the Texas City distillation tower overfill. The current cases indicate that in the age of automation, reliable FDD algorithms should be designed to effectively identify sensor signals rather than allowing them to be misread or drowned out by the volume of data [7]. For example, in the BP Toledo Refinery incident, the alignment of valves and pumps deviated from the predetermined position, but the fault detection system failed to flag this fault through context-aware marking until a flammable gas release occurred.

From the past Industry 3.0 era to the current Industry 4.0 era, the science of FDD in process systems has been constantly updated and

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Nomenclature

Acc	Accuracy
AE	Autoencoders
AI	Artificial intelligence
BN	Bayesian networks
CNN	Convolutional neural networks
CPT	Conditional probability table
CSB	Chemical Safety and Hazard Investigation Board
CSTR	Continuous stirred tank reactor
DBN	Deep belief networks
DL	Deep learning
DT	Digital twin
FDD	Fault detection and diagnosis
FDR	Fault detection rate
FPR	False positive rate
F1	F1-score
GNN	graph neural networks
KPCA	Kernel principal component analysis
LIME	Local interpretable model-agnostic explanations
LLM	Large language model
LSTM	Long short-term memory
ML	Machine learning
PCA	Principal component analysis
PICO	Population, Intervention, Comparison and Outcomes principles
PRISMA	Preferred reporting items for systematic reviews and meta-analysis
Pr	Precision
SHAP	Shapley additive explanations
SPE	Square prediction error
SVM	Support vector machine
TEP	Tennessee Eastman process
T^2	Hotelling T^2
t-SNE	t-stochastic neighbor embedding
XAI	Explainable artificial intelligence
XGBoost	Extreme gradient boost
DCS	Distributed control systems

Table 1
Foundational loss events shaping early FDD development.

Year	Facility (Country)	Process unit context	Immediate technical cause
1974	Nypro, Flixborough (UK)	Cyclohexane loop	Pipe bypass ruptured
1984	Union Carbide, Bhopal (India)	MIC storage	Relief valve lift
1988	Piper Alpha (UK)	Gas condensate export	Condensate line opened after maintenance
1989	Phillips 66, Pasadena (USA)	Polyethylene reactor	Air separation opened
2005	BP, Texas City (USA)	Hydrocarbon unit	Level transmitter stuck

Table 2
Recent loss events reflecting current challenges in data-driven FDD.

Year	Facility (Country)	Process unit context	Immediate technical cause
2019	PES Refinery (USA)	HF alkylation unit	Corroded pipe elbow ruptured
2019	KMCO, Crosby (USA)	Isobutylene sulfuration	Fatigued piping fractured
2021	LyondellBasell, La Porte (USA)	Acetic acid recovery	Valve disassembled under pressure
2022	BP Toledo Refinery (USA)	Crude distillation	Valve system misalignment
2023	East Palestine derailment (USA)	Vinyl chloride tank	Delayed thermal detection

iterated, as shown in Fig. 1. To expand our knowledge as much as possible, we outlined multiple key points in the evolution process based on four pillars: **data acquisition, signal preprocessing, condition diagnosis**, and **predictive maintenance**. Each ring represents a progressive stage of technological maturity, moving outward from traditional instrumentation to modern AI-augmented frameworks. Within the first quadrant, early industrial systems are dominated by hard-wired programmable logic controllers, whose limited storage capacity confined datasets to a narrow range of variables. The subsequent adoption of smart sensors enabled the capture of unstructured operational data, resulting in higher-fidelity and more continuous data streams. The emergence of wireless sensor networks and edge artificial intelligence (AI) devices expanded spatial coverage, which enabled the collection of operation status data tagged with contextual information and laid the foundation for later preprocessing stages. In the second quadrant, inner-ring practices focus on offline spreadsheet scripts, generating single-purpose metrics interpreted by experts. Large language model (LLM)-driven signal and context integration, such as combining sensor feedback and P&ID text to automatically generate enriched feature tensors, will become the future focus of evolution.

In the condition monitoring and diagnosis section, reference thresholds, alarm matrices, and first-generation expert systems are sufficient when variable coupling is low. With the progress of industrial digitalization, data-driven classifiers such as support vector machines (SVMs), convolutional neural networks (CNNs), and long short-term memory (LSTM) structures are used to decode multivariate features. In the Industry 4.0 frontier, transformer structures, physical information neural networks, and digital twin (DT) [8] produce explainable root cause heatmaps while generative AI can interact with human operators to provide possible fault narratives and recommend verification tests. In the final quadrant, looking outward, reliability-centered predictive maintenance and health management frameworks incorporate Bayesian networks to update remaining useful life [9]. At the cutting edge of this quadrant, federated learning on degradation data across the entire system enables training curves in different scenarios and transferring them to a centralized model for decision-making.

To sum up, the above arrangement visualizes not only the historical development of FDD technologies but also the integrated pathway leading toward fully data-driven, intelligent, and interpretable process fault diagnosis systems. Among these, progress in one quadrant necessarily promotes advances in adjacent quadrants. For example, preprocessed data with multivariate features enhances the reliability of diagnosis, which drives maintenance actions in the predictive quadrant to achieve information closure, thereby influencing process operating mechanisms and the generation of future data. From a technical trajectory viewpoint, an advanced systematic literature review framework is proposed for a comprehensive overview of the literature related to FDD in industrial process systems. Through analysis and exploration of the questions formulated according to the Population, Intervention, Comparison, Outcomes (PICO) principles, the current trends in industrial process fault diagnosis are clarified, and potential directions are suggested based on the existing research gaps.

2. Methodology

During the systematic examination of current research for industrial systems, the term methodology refers not only to retrieval strings and

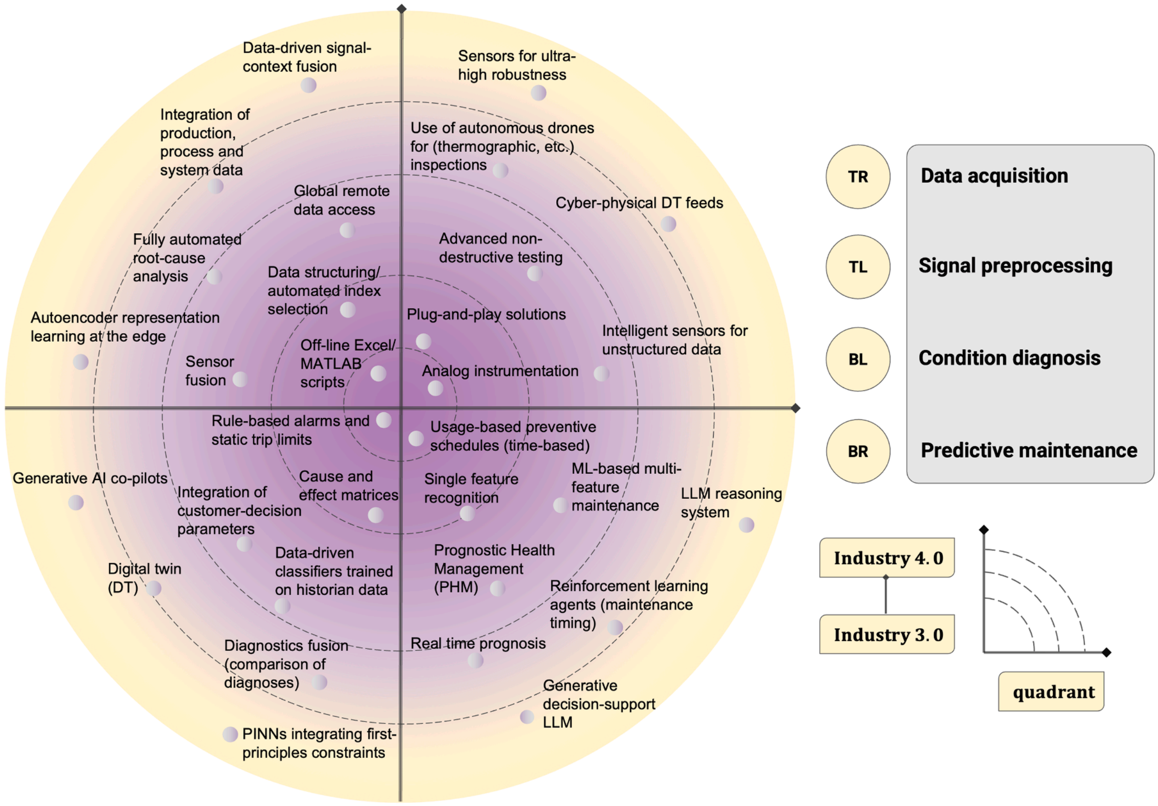

Fig. 1. The technical trajectory of industrial process fault detection and diagnosis.

statistics, but also to the knowledge boundaries covered by review itself. In this study, a rigorous structured literature review method, the set of Preferred Reporting Items for Systematic Reviews and Meta-Analysis [10] (PRISMA) guidelines, is adopted for strategic investigation, which involves analysis of the research subjects and exploration of hot topics. Based on the 11-item method checklist in the latest PRISMA [11, 12] statement, the review framework for data-driven FDD is proposed and divided into four stages: review initialization, inclusion and exclusion strategies, quality appraisal scales, and data synthesis. Together, these stages provide a systematic pathway from literature identification to synthesis, ensuring that every inclusion decision is explicitly documented. It is worth noting that the proposed structure can facilitate future comparative studies, allowing subsequent reviews to extend, refine, or replicate the present findings with confidence.

2.1. Review initialization

In the initialization step, ScienceDirect, Web of Science Core Collection, and IEEE Xplore are selected as the data sources for literature selection, which cover most of the innovative research in the field of engineering. Given the differences in the advanced functions of various databases, distinct Boolean strings are used for article retrieval. For example, ScienceDirect lacks the proximity operator but supports the high-precision TITLE-ABS-KEY() tag, while Web of Science offers the flexible TS=() subject search functionality and allows on keyword constraints within a specified range. The optimized strings, shown in Table 3, significantly reduced the time required for further exclusion of irrelevant hits compared with a general keyword search.

2.2. Inclusion and exclusion strategies

In Table 4, independent inclusion and exclusion strategies are defined to preliminarily clarify the scope of this study, ensuring maximum relevance while minimizing bias. The time scope cutoff is set in 2015 to focus on the most recent decade in which data-driven and deep learning (DL) methods became viable technologies in industrial process systems. Although previous studies are pioneering, their

Table 3
Boolean retrieve strings for targeted databases.

Database	Syntax note	Boolean retrieve strings
ScienceDirect	TITLE-ABS-KEY() only	TITLE-ABS-KEY(("fault diagnosis" OR "condition monitoring" OR "remaining useful life prediction" OR "anomaly detection") AND ("industrial process" OR "chemical process" OR "process system"))
Web of Science Core Collection	Topic search (TS=) with proximity	TS=(("fault diagnosis" OR "condition monitoring" OR "remaining useful life prediction" OR "anomaly detection") AND ("industrial process" OR "chemical process" OR "process system"))
IEEE Xplore	Combine field tags	(("Document Title": "fault diagnosis" OR "condition monitoring" OR "remaining useful life prediction" OR "anomaly detection") AND ("Abstract": "industrial process" OR "chemical process" OR "process system"))

Table 4
Inclusion and exclusion strategies.

Domain	Inclusion strategies	Exclusion strategies
Time scope	Over the past decade (2015-now)	Published before 2015
Language	English	Other languages
Publication type	Peer-reviewed journals	Abstract only, conference paper and poster, book chapters
Focus of interest	Industrial processes	Mechanical or computer systems

findings have been adequately covered in existing reviews [13,14], so excluding these can sharpen the technological perspective without obscuring the historical context. Restricting the screening repository to English language articles ensures consistency in technical terminology and avoids potential translation errors when comparing statistical metrics across multiple language sources. Additionally, limiting the scope to peer reviewed journals filters out conference abstracts, which often omit hyperparameter settings, thus enhancing methodological reproducibility. Finally, the focus is explicitly placed on industrial process systems, excluding purely mechanical or computer science-oriented fault diagnosis papers.

2.3. Quality appraisal scales

To ensure fair and consistent evaluation, the quality appraisal scales are developed to focus on industrial scenarios, data-driven methods, relevant benchmark evaluations, and rigorous validation, aligning PRISMA principles with the needs of fault diagnosis domain. Therefore, after applying the exclusion strategies, we conducted further structured filtering through a careful review of the abstract, methodology, and conclusion sections, which is lacking in current systematic studies [15, 16]. The four scientific quality appraisal scales are defined and outlined below:

1) Is the main topic of this study focused on multi-variable systems in industrial processes?
2) Is the method presented in this paper essentially a data-driven model or algorithm?
3) Is the complete fault diagnosis pipeline evaluated for reliability on relevant real signals or simulated datasets?
4) Is the algorithm validated by a rigorous procedure that includes cross-validation, independent testing sets, or repeated runs?

As shown in Fig. 2, the quality appraisal scales are scientifically proposed based on four aspects. From a multivariate industrial perspective, the target paper is required to mention specific industrial units (such as distillation towers, catalytic cracking units, or bioreactors) and list multiple measurement variables or control loops. From a data-driven innovation view, the target paper is expected to claim a novel or modified data-driven algorithm, which include attention-enhanced LSTM, graph autoencoders (AE), and sparse principal component analysis (PCA). From a model evaluation viewpoint, the target paper is intended to use actual plant data and existing benchmark datasets as references. When considering training reliability, rigorous validation procedures should be employed to ensure fair comparison and reproducibility. Manuscripts that do not meet the four criteria above are all excluded.

2.4. Data synthesis and implementation

The comprehensive procedure for transforming the initial database findings into an evidence repository for FDD reviews of industrial process systems is presented in Fig. 3. In the initial steps, the identification process primarily relied on the ScienceDirect, Web of Science Core Collection, and IEEE Xplore databases, yielding preliminary metadata counts of 646, 995, and 180, respectively. Prior to full-text screening, 388 duplicate entries were removed through association analysis to limit downstream bias in the citation list. Specifically, 236 items were flagged due to overall similarity scores $\geq$95 %, and 152 were marked by syntactic title overlap. In the subsequent screening phase, we employed the inclusion and exclusion matrix described in Section 3.2, where 701 records were filtered out at the abstract level among 1433 available records. Most of the removed records (n = 436) fall outside the 2015 - 2025 publication window, and some articles (n = 244) fail to comply with the research type requirement.

In further quality appraisal, we carefully inspected the full texts to confirm four elements: industrial process issues, data-driven methodology, reliability of training, and validation against relevant benchmarks. This rigorous step additionally excluded 655 ineligible papers, and 5 extra papers were discarded through the second round of similarity checks. The 72 retained articles constitute a strictly screened dataset that balances methodological accuracy and evidence robustness. Importantly, all literature were organized into a categorized directory in Zotero for responding to critical research questions and discussing review findings.

2.5. Research question formulation

Each research question is formulated based on the Population, Intervention, Comparison, Outcomes principles [17,18] to ensure consistency with the previous search strategies and screening rules. The population refers to the target group of the study, industrial process systems. An intervention refers to the specific FDD technology under review, including the proposed framework and methodology. The

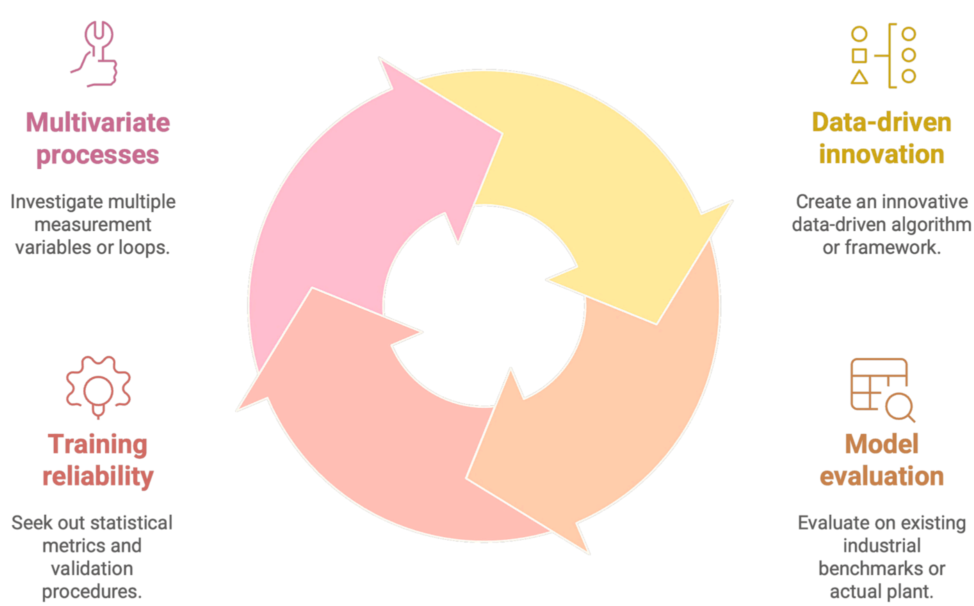

Fig. 2. Four aspects of quality appraisal scales.

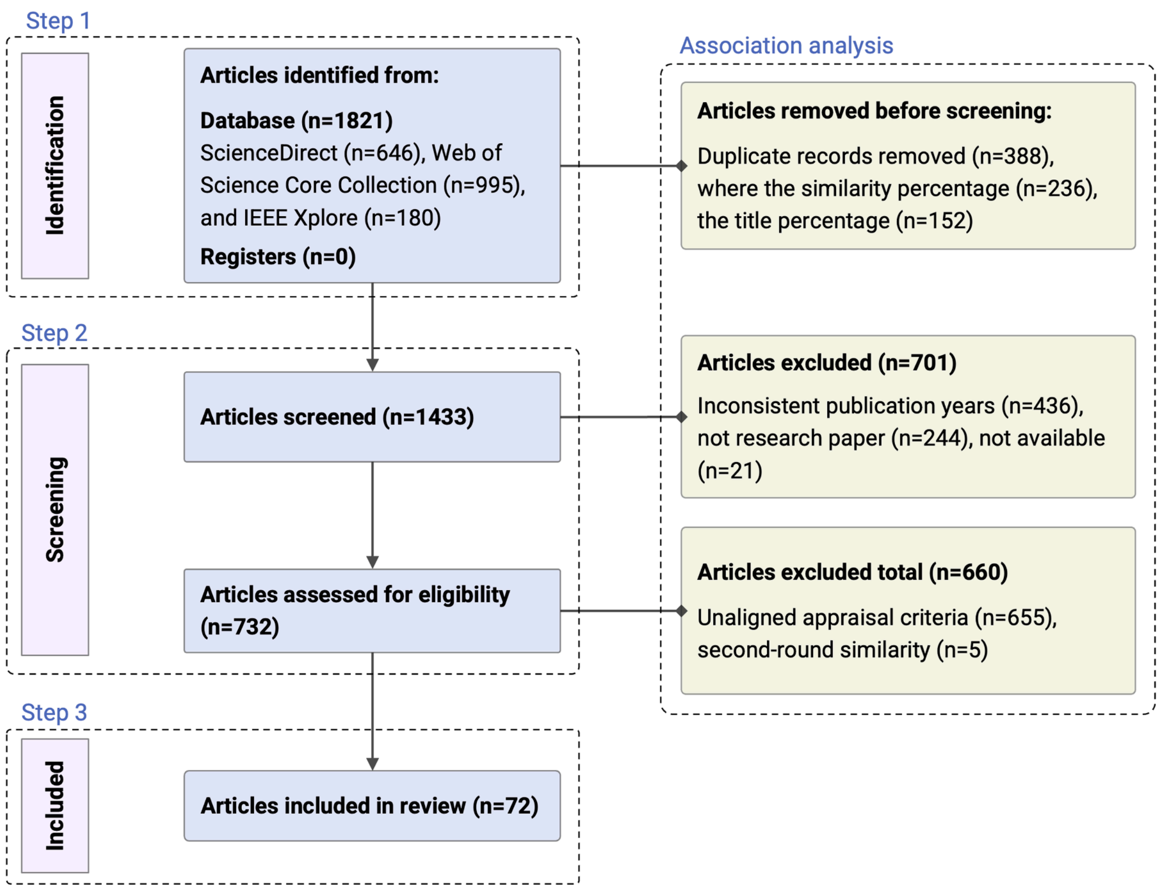

Fig. 3. A comprehensive overview of PRISMA evidence repository acquisition for industrial process FDD.

comparison [19] specifies the benchmark for evaluating the intervention. The outcomes list the evaluation results corresponding to the intervention, including accuracy (Acc), fault detection rate (FDR), and F1-score (F1). As summarized in Table 5, this study adheres to four research objectives. These proposed inquiries ensure alignment between initial database retrieval, inclusion criteria, and evidence synthesis while enhancing the clarity of comparative analysis across various FDD methodologies. It is worth noting that with the PICO cells populated, Boolean retrieve strings reflecting each element vocabulary are drafted.

3. Review results and discussion

3.1. Current advances in industrial fault diagnosis

Fig. 4 displays both annual and cumulative publication counts of fault diagnosis studies that satisfied the strict PRISMA screening rules outlined in the previous section. Between 2015 and 2019, fewer than eight qualified papers per year appeared in industrial process systems, reflecting the dominance of traditional multivariate statistical control methods and the limited availability of open benchmarks. A significant inflection occurred in 2021, triggering the acceleration window marked in the chart, which can be attributed largely to the maturation of open-source DL frameworks that lower the barrier to entry.

Additionally, the themes of the screening literature are concentrated around fault diagnosis, closely adjacent to DL and its classic form, convolutional neural networks, which aligns with the growth in the number of deep learning papers [20–22]. Focus should be given to the causal intervention, which is consistent with current fault diagnosis efforts that emphasize physics-guided and explainable algorithms. Together, these findings collectively form the complete background for the P element of the PICO principle, i.e., the current progress in industrial process fault diagnosis. The rest of this paper shifts to exploring specific intervention measures, as well as the outcomes attained in terms of diagnostic accuracy and robustness.

Table 5
Research objectives based on PICO principles.

ID	PICO element	Research question
RQ_1	Population	What is the current landscape of scientific research in this field?
RQ_2	Intervention	What FDD techniques involving data-driven interventions have been proposed for industrial process systems, and what distinctive algorithmic innovations characterize each approach?
RQ_3	Comparison	Which data benchmarks are implemented to compare the current intervention measures?
RQ_4	Outcomes	What are the main statistical performance metrics for industrial process fault diagnosis?

3.2. Algorithmic interventions in process systems

Early studies emphasized knowledge-driven approaches such as expert systems [23,24], fuzzy logic controllers [25] and evolutionary algorithms [26], which encoded operator experience and heuristic reasoning to handle nonlinearities and uncertainty in process data. However, under the PRISMA screening criteria employed in this work, these paradigms are not categorized as independent families because their mechanisms primarily rely on rule-based inference or optimization heuristics rather than data-driven learning. The subsequent subsections therefore focus on five dominant data-driven families that collectively represent the current state of algorithmic interventions in process fault diagnosis.

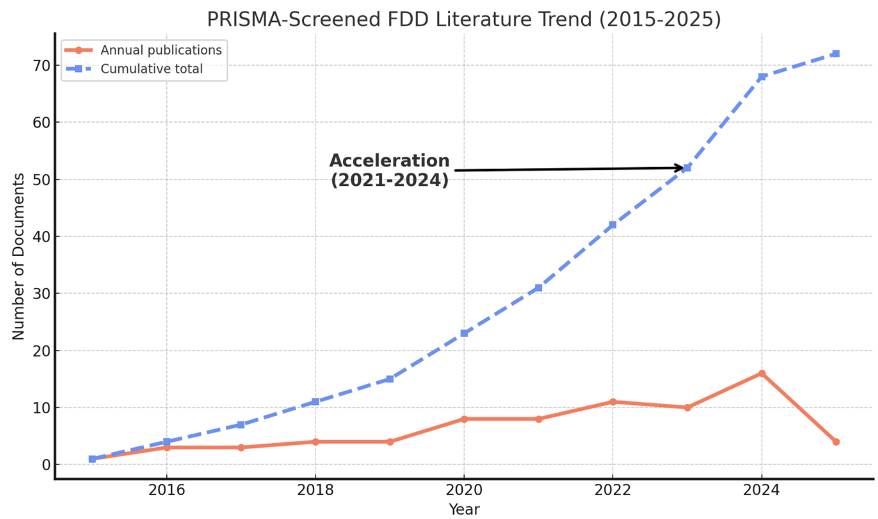

Fig. 4. Temporal distribution of PRISMA-screened fault diagnosis publications.

3.2.1. *Classification of data-driven algorithms*

The FDD research in industrial process systems has expanded from a handful of multivariate statistical techniques in the early 1990s to hundreds of heterogeneous algorithms today [27]. To facilitate systematic review and address the target questions, a five-family classification framework is proposed to generalize the methods encountered in the screened literature into five mechanism-led groups. Each family is characterized by the core principle used to generate diagnostic statistics, and the classification scheme is visualized in Fig. 5. The broad arrow on the left represents the complete collection of data-driven fault diagnosis methodologies retrieved by the PRISMA protocol. The gray pyramid symbolizes the high-dimensional feature derived from historical plant data or first-principles simulation, and candidate algorithms are sampled in the feature space to capture fault information. Consequently, the above proposed families are mechanism-oriented and not mutually exclusive. Each paper is assigned to the group whose primary innovation generates the diagnostic signal (e.g., transfer/domain adaptation under distribution shift), even when the base classifier is statistical or deep learning.

The statistical method indicated by the uppermost arrow relies on linear or kernel projection to define control limits around normal operating status. Principal component analysis [28], local outlier factor [29], and linear discriminant analysis [30] are commonly employed to perform industrial process monitoring and fault detection tasks. Bayesian networks (BN) interpret problems as probabilistic patterns, using graphical models and Gaussian processes [31] to propagate measurement uncertainty and prior knowledge. Deep learning architectures make use of hierarchical representation learning [32–34], incorporating convolutional, recurrent, or attention mechanisms to identify features that can be classified as faults. Transfer learning recycles knowledge from related factories or process conditions [35,36], leveraging domain adaptation to mitigate the long-term shortage of labelled fault data in industry. The bottom ensemble component highlights various foundational methods [37], including conventional machine learning approaches, explainable ensembles, and a few hybrid diagnostic algorithms.

3.2.2. *Statistical method*

Statistical methods have been some of the most widely used approaches for industrial FDD due to their balance between

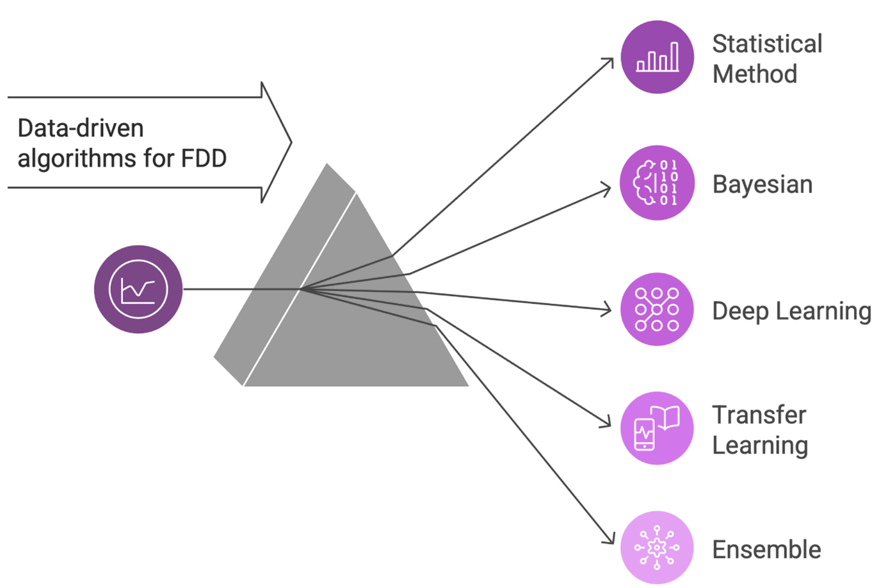

Fig. 5. The proposed methodological classification of data-driven algorithms for fault detection and diagnosis. Families indicate mechanism-oriented clusters derived from PRISMA-screened literature; conceptual overlap may exist, but each paper is grouped according to its dominant diagnostic mechanism.

interpretability and computational efficiency. Fig. 6 outlines a generalized flowchart of statistical methods for fault detection and diagnosis in industrial systems. The top left panel describes the problem, where simulated measurements from a chemical stirred reactor are continuously recorded and stored in a historical database. Since the raw variables differ in units and dynamic range, the first step is autoscaling and mean-centering. This unit variance transformation can eliminate magnitude disparities, ensuring that all variables contribute equally to downstream feature extraction. The algorithm core module on the right, exemplified by PCA, represents the projection of the raw variable latent space onto a low-dimensional score space via a loading matrix [28,38]. In this case, the ellipses indicate the multivariate confidence region, and the clustering separation of normal and fault states confirms the ability of PCA to learn latent features. Finally, the bottom left evaluation panel reports two key components: (i) a comparison of various algorithms on squared prediction error (SPE) and Hotelling T^2 (T^2) statistical metrics [29], which are discussed in detail in Section 3.4; and (ii) a labeled timeline strip that exposes the temporal patterns of predicted classes. Collectively, these elements provide a thorough perspective for the following discussion, capturing how classical statistical methods monitor process states and diagnosis deviations.

However, the performance of vanilla PCA deteriorates when nonlinear effects dominate. To address this issue, kernel principal component analysis (KPCA) extends PCA to a high-dimensional feature space using kernel functions, effectively capturing complex nonlinear interactions among variables. In recent studies, Zhang et al. [39] introduced a dynamic KPCA algorithm that decreases the matrix computation dimension via periodic splitting and merging. When applied to continuous and intermittent process benchmarks, the proposed model reduces the average false alarm rate by filtering predictable autocorrelated signals while enhancing the detection capability of slowly evolving fault root causes. In [40], Sun et al. establish a dynamic regression framework for distributed KPCA that can locate fault variables with only one graph and exhibit higher fault detection rates with respect to partial least squares. Although multivariate statistical control algorithms remain the bedrock of industrial monitoring, their sensitivity to non-stationarity inevitably limits coverage in highly integrated systems [41], motivating a layered progression of methods explored in the subsections that follow.

3.2.3. *Bayesian networks*

As depicted in Fig. 7, a three-phase flow case is selected to integrate the screening research based on BN into four visual tiers, consistent with the pipeline mapping in the previous section. Assume that the problem context is that multiple feed streams converge at a catalytic column, and the raw fault information sequence of the sensor array is imported into the dataset. Subsequently, continuous variables need to be discretized to meet the requirements of the conditional probability table (CPT) of BN [42,43]. Here, the blue panel displays three discretization strategies: equal-width, entropy-based, and a mixed raw distribution, where the choice of strategy affects the posterior sensitivity. The right-hand side shows the static Bayesian slice (top) and its dynamic unfolding (bottom). The gray circle is an unobservable potential fault state, the blue circles denote sensor variables, and the green nodes represent derived quality indices. A row of alarm nodes transforms the posterior probability into a binary decision through an adjustable risk threshold. Finally, the multivariate runtime chart of posterior means and the maintenance time diagram with confidence bands [44] provide a basis for predictive fault detection.

In process monitoring tasks, dynamic BNs are used to model the evolution of sensor distributions over time. The posterior prediction check is updated by integrating statistical methods [45], and sensor malfunctions can therefore be considered according to indicators expressed in the form of log-likelihood residuals. An application to an actual wastewater treatment system suggests that [46], with respect to probabilistic PCA, the proposed BN-based method can automatically select the dimension of the latent space and results in a lower false negative rate. Liu et al. [47] proposed a multi-state BN optimization approach that clarifies the causal relationships between internal variables via transfer entropy, thereby filling the gap in traditional statistical

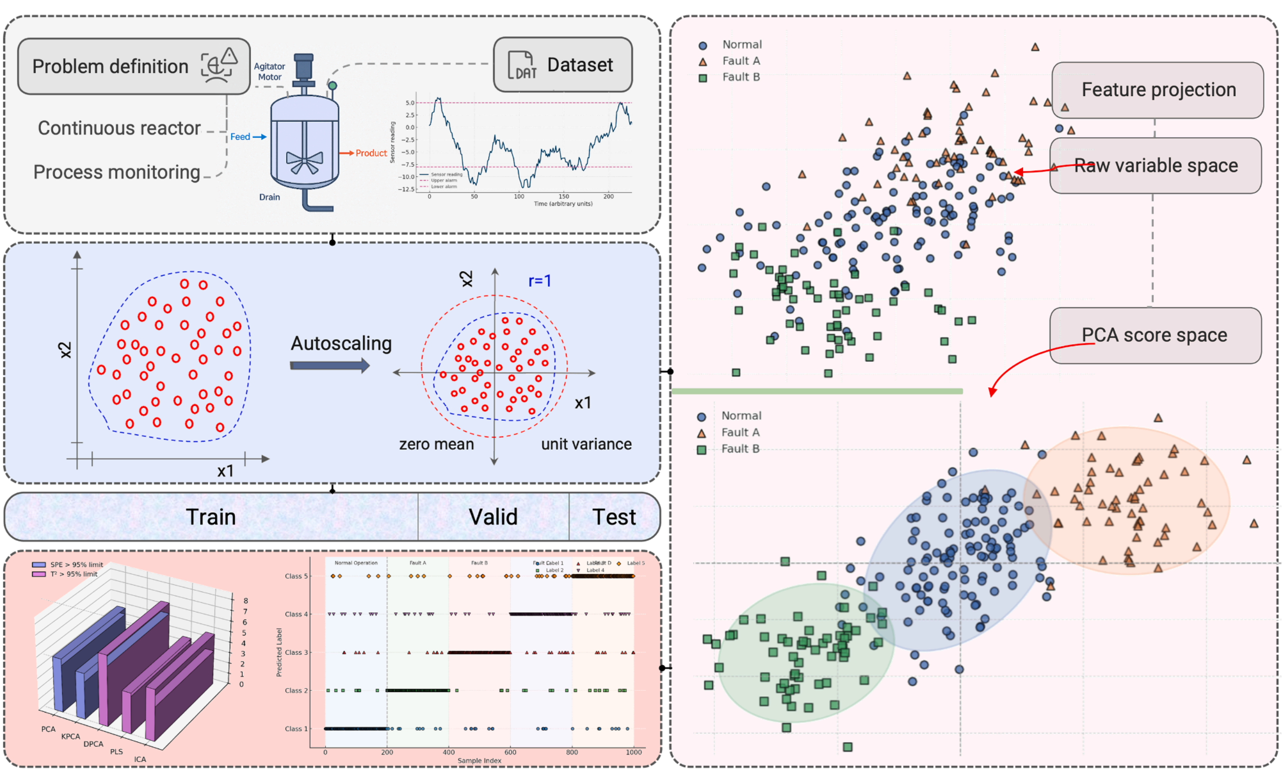

Fig. 6. Generalized pipeline for fault detection and diagnosis with statistical method.

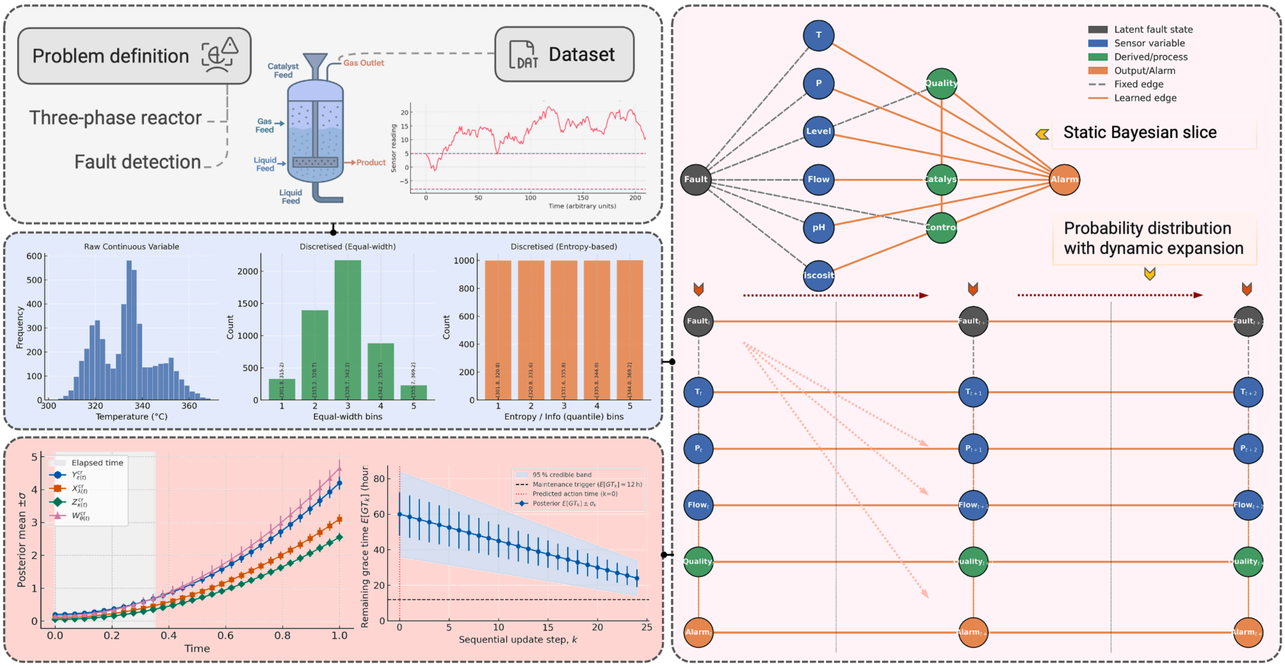

Fig. 7. Generalized pipeline for fault detection and diagnosis with Bayesian networks.

methods regarding fault propagation path identification.

Furthermore, Bayesian ideas commonly demonstrate a high degree of proficiency in causal reasoning [48], allowing for fault classification via maximum posterior inference once CPT training is complete. In Yu and Zhao's work [42], a probabilistic ensemble learning BN framework is constructed to return the a posteriori distribution of fault categories in response to new evidence vectors. Basha et al. [31] proposed a logic-based BN decision model incorporating Gaussian processes, which has better fault diagnosis accuracy than improved PCA. However, the current challenge lies in the exponential growth of real-time inference efficiency as network complexity increases or dataset size scales up. Combining the propagation uncertainty and causal analysis capabilities of BN with the advantages of DL, which is discussed in the next subsection, can serve as a reliable solution for process monitoring.

3.2.4. *Deep learning*

The typical workflow adopted in deep learning studies is synthesized into a comparable four-stage pipeline shown in Fig. 8. The upper left block describes the research problem: fault diagnosis by processing historical data from a continuous reactor. Afterward, the raw trends are divided into segments using a sliding time window, which captures the dominant process dynamics by adjusting the stride. The resulting three-dimensional tensor is then split into training, validation, and test folds to satisfy cross-validation requirements. The upper right panel depicts the algorithm core, here exemplified by a bidirectional LSTM network augmented with an attention layer that weights temporally salient features [49]. The bottom right quadrant summarizes the evaluation results, where a confusion matrix visualizes the category detection ability, and bar statistics present several performance metrics. It is worth noting that although LSTM is highlighted, the same pipeline remains applicable

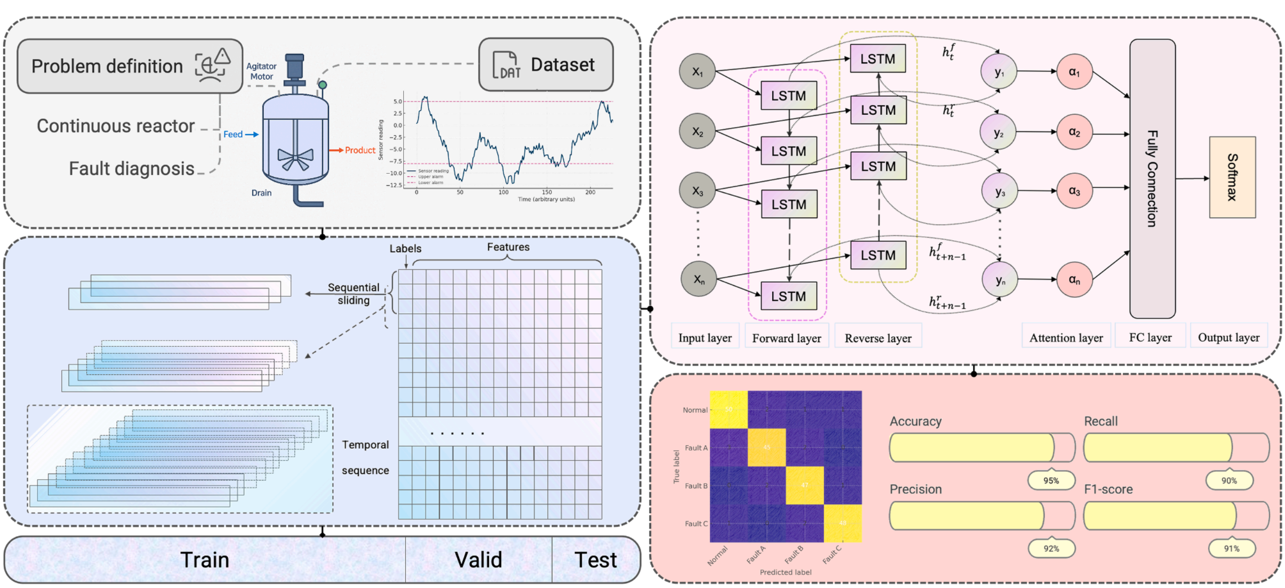

Fig. 8. Generalized pipeline for fault detection and diagnosis with deep learning.

to CNN and AE by swapping the core block.

Early DL attempts in industrial FDD exploited deep belief networks (DBNs) [21] to learn fault features from the spatial-temporal domain. When applied to the Tennessee Eastman process (TEP), one such DBN achieved [50] an average fault detection rate 7.5 % higher than PCA methods. Building on this foundation, Wu and Zhao [51] proposed a multilayer CNN algorithm that further improved the average detection rate of this chemical process by 6 %, reducing both parameters and computational costs. As research deepened, a process topology convolutional framework was presented in [52] that can efficiently identify system faults through physical principle information. Tao et al. [53] constructed a triage-based CNN that enables timely fault warning by dividing faults into distinct sets and performing diagnosis. Moreover, in [54,55], and [56], the original input was converted into a matrix graph to enhance the reliability of the diagnostic pipeline by utilizing the image recognition capabilities of CNN.

Despite CNNs excelling at local pattern extraction, they struggle with the long-range dependencies inherent in slow chemical kinetics. LSTM overcomes this limitation by maintaining forward or inverse hidden states that encapsulate past and future context within the sliding time window. In [57], a pipeline combining CNN and LSTM is proposed for industrial fault forecasting, which can recognize multivariate faults one sampling step in advance. Huang et al. [58] and Ren et al. [59] integrated similar workflows for process fault diagnosis, the former introducing Gaussian white noise into the benchmark to enhance the noise resistance of the model, and the latter compressing the output into hyper features space for semi-supervised learning. Additionally, attention modules can be stacked on the connected hidden vectors (in Fig. 8) to focus the classifier on the most informative time slices, as demonstrated in the work of Xiong et al. [60] and Zhao et al. [61].

Autoencoders occupy a middle ground between purely discriminative CNN/LSTM classifiers and statistical variable models. A standard AE compresses each input window into a low-dimensional latent vector for reconstruction, and the reconstruction error can be interpreted as a statistical measure SPE [20,62], with values exceeding a threshold indicating anomalies. Cacciarelli and Kulahci [63] designed a variational AE by configuring a regularization term in the loss function, making it possible to determine the root cause of industrial process disturbances. In recent studies, Han et al. and Jing et al. combined discriminative methods with AE to propose a composite fault detection framework, where [64] achieved a detection rate of 98 % in an industrial distillation column system, while the other [65] was able to classify faults with only 25 % of labeled data. In summary, CNN, LSTM, and AE series each contribute distinct strengths to data-driven fault diagnosis, including spatial feature extraction, temporal context information retention, and unsupervised anomaly scoring.

3.2.5. *Transfer learning*

Building upon deep learning architectures, transfer learning extends model utility across different industrial domains by embedding cross-context generalization as the main diagnostic mechanism. The representative pipeline illustrated in Fig. 9 captures how transfer learning is operationalized in current FDD frameworks. The gray panel contrasts the high frequency data streams of the source domain and the target domain, which exhibit differences in amplitude and variance. The temporal structure is then visualized in the light blue module, where the pink dots denote raw feature embeddings of the source signals, and the green dots represent target features projected into the same latent space. Typically, before feature mapping is performed, the distribution does not fulfill the domain invariance prerequisite for classifier transfer [35, 66]. The right block depicts the dual encoder architecture, with the upper path optimized on labeled source data while sharing the initial convolutional layers with the lower purple path. Minimizing the adversarial loss is achieved by adjusting the deeper dense blocks, forcing the hidden representations to be indistinguishable between domains. Lastly, the t-SNE plot shows multiple fault clusters in the adapted latent space [67], while the violin plot compares the statistical metric across multiple baseline classifiers [36], highlighting the benefits of fine-tuned transfer learning. From a mechanism-oriented perspective, the diagnostic signal in transfer learning-based studies arises from the degree of feature alignment achieved between domains, which directly governs the model's capacity to distinguish process deviations in new operating environments. Accordingly, transfer learning is classified here as a distinct mechanism family, not due to the algorithmic isolation, but because it defines a unique causal pathway of fault generalization that can bridge heterogeneous process conditions and enhance industrial applicability.

The exploration of transfer learning in fault diagnosis starts with attempts to transfer statistical algorithms to target tasks, such as in [36], which recalibrated linear discriminant analysis via eigenvalue weights to detect disturbances in a hydrocracking system. However, the shortcomings of this approach became apparent [68] when faced with

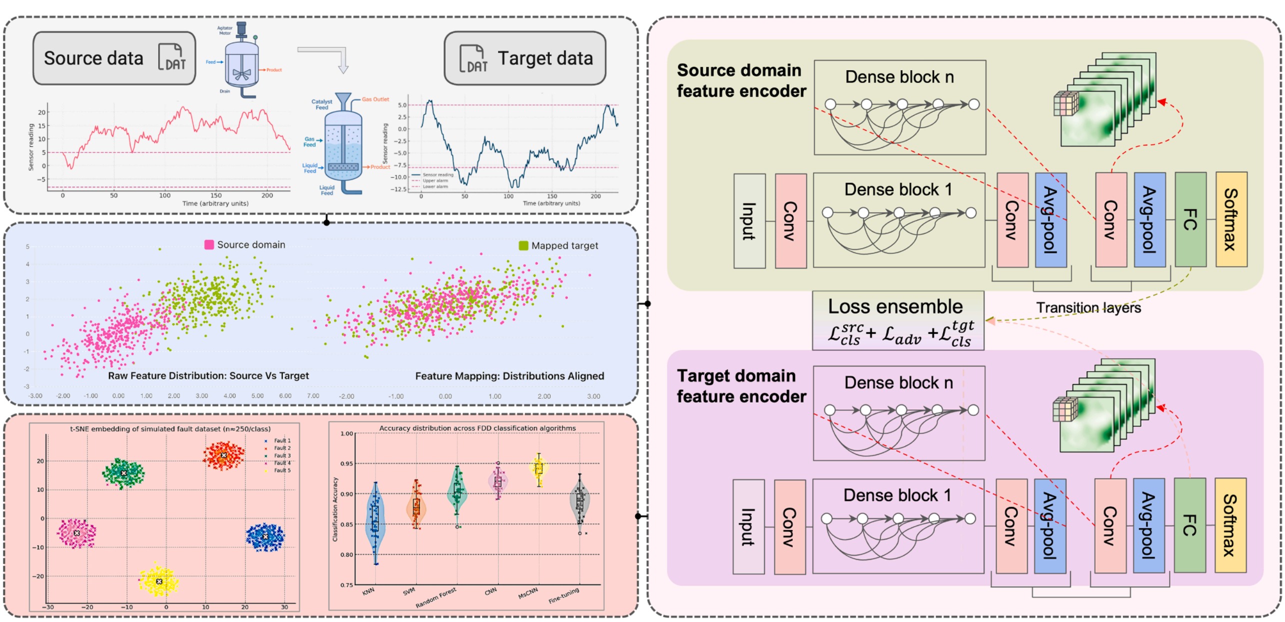

Fig. 9. Generalized pipeline for fault detection and diagnosis with transfer learning.

nonlinear changes in the form of catalyst deactivation or sensor drift. Deep transfer learning mitigates these vulnerabilities by learning domain-invariant features directly from the raw data dimension. Deng et al. [69] developed a cross-domain learning strategy based on stacked AE for fault diagnosis in wind energy systems, demonstrating transfer performance across multiple experiments through box plots. Transformer architectures employ multi-head self-attention with positional encoding to capture long-range temporal dependencies and cross-variable interactions without recurrence, enabling parallel sequence modeling on multivariate process streams [7]. Qin et al. integrated multiscale CNN and Transformer as domain feature extractors [70], aligning the joint distributions of several domains while performing fault diagnosis of multimodal chemical process problems. Xiao et al. [71] utilized Transformer models to analyze key features in discrete time series signal and enhanced them through transfer learning strategies. Furthermore, meta-learning can treat disparate chemical functional units as sub-targets for rapid optimization, and Gao et al. [72] have validated an enhanced domain generalization method across three industrial process benchmarks. In future studies, emphasis should be placed on interpretable transfer mechanisms that integrate domain knowledge with deep embeddings, clarifying why specific feature mappings reveal hidden fault behaviors [73,74]. This interpretive focus will determine whether transfer learning evolves into a core component of next generation industrial fault detection, complementing ensemble and XAI paradigms discussed in the following section.

3.2.6. Ensemble

Single model detectors emphasize achieving high specificity within certain assumptions, while ensemble frameworks achieve robustness by aggregating complementary learners. Through a thorough review of relevant literature, three dominant architectures are revealed, consisting of classical machine learning (ML) ensembles, hybrid statistical and deep learning methods, and explainable ensembles. In the first category, Quatrini et al. [75] established an industrial anomaly detection framework integrating decision forests and decision jungles, which can achieve a Precision and FDR above 0.99 for 16 process parameters in a fluidized bed system. Given the poor performance of the Softmax algorithm in complex industrial conditions, Chen et al. [76] introduced an XGBoost method into the diagnostic framework to replace the classification layer of neural networks, gaining an additional one percentage point in F1-score while preserving inference latency suitable for real-time control.

Hybrid statistical and DL methods combine explainable statistical monitoring layers with black-box neural networks to integrate latent variable models rather than replace them. In [77], non-negative matrix factorization and PCA are applied as raw variable transformations, enabling enhanced LSTM with feature reduction to achieve an FDR of 0.985 in a fast sampling industrial system. Ali et al. [78] integrated PCA, Shannon information entropy, and wavelet transform to design a novel fault identification theory that can explain the propagation path of the TEP benchmark with a lower fault alarm rate. However, the multi-stage stacking involves significant computational overhead for hyperparameter tuning, and the hard voting strategy of DL ensembles can ignore critical alerts.

In the last type of ensemble approaches, research can be performed based on explainable artificial intelligence (XAI) to generate local or global interpretations to assist system fault diagnosis. For instance, Harinarayan and Shalinie [79] developed a fault detection and correction mechanism using the TreeSHAP algorithm, which can utilize factual explanations as a solution for Eastman process fault diagnosis. Another explainable ensemble pipeline integrates graph neural networks (GNN) [80], which encode process topology and sensor interconnections through node-edge relationships to enhance multivariate dependency learning. As an example, Liu et al. [81] employed mutual information selection of GNN guided by causal theory to identify root cause variables and improve diagnostic precision. Such graph-based reasoning allows models to represent spatial and structural correlations across process units - particularly useful in distributed systems like pipelines [82] or heat-exchanger networks, where faults propagate along physical connections. In [83], a gradient-enhanced causal learning ensemble is presented to extract fault information under the Pareto optimality of two objectives. The three-phase flow benchmark verification shows that proposed explainable framework can improve the predictive and causal correlation with an FDR of 0.984 or higher. Considering the significant differences among the various types of ensemble architectures, this section does not provide a generalized fault diagnosis pipeline.

3.2.7. Comparative overview of algorithmic families

To consolidate our findings from Sections 3.2.1–3.2.6, Table 6 provides a high-level overview of the dominant algorithmic families applied in industrial fault detection and diagnosis. Each family demonstrates distinct trade-offs between interpretability, data dependency and implementation feasibility, reflecting the diverse operational constraints of real process systems. Statistical and Bayesian models continue to dominate in existing plants due to their simplicity and auditability, whereas DL and ensemble architectures demonstrate superior performance in complex, nonlinear environments when sufficient labeled data and computational capacity are available. Transfer learning provides an intermediate pathway for plants operating across heterogeneous modes or with limited labeled data, offering improved adaptability when domain discrepancies are moderate. Collectively, these insights signal a steady evolution from rule-based detection toward adaptive and explainable paradigms, which are further elaborated in Section 4.1 on scenario-dependent applicability gaps. The coverage of fault diagnosis, condition monitoring, and predictive maintenance throughout the reviewed studies is provided in Appendix (Table A1) to enhance structural transparency.

3.3. Data benchmarks for reliability assessment

Data benchmarks occupy the same role in fault detection and diagnosis research that contextual information does in LLM, which lets disparate studies speak a common quantitative language. In the early stages of the study of FDD in industrial processes, researchers primarily relied on linear, single-loop cases, which are essentially teaching models used to validate classical statistical process control. This paradigm shifted in the mid-1990s when Downs and Vogel [84] introduced the TEP simulator, which incorporated 41 continuous measured variables, interacting heat and mass transfer kinetics, and 21 predefined fault scenarios. Since its release, the TEP has become the most widely used reference model for evaluating diagnostic reliability, largely because real industrial data remain difficult to access due to confidentiality and safety constraints. As a result, simulation-based benchmarks such as TEP continue to serve as essential reproducible environments for assessing algorithmic performance and enabling fair comparison across studies.

The second milestone is the publication of the continuous stirred tank reactor (CSTR) simulator by [85] and [86]. Although its dimensionality is much smaller than that of the TEP benchmark, the CSTR remains highly regarded because it exhibits rigid nonlinear dynamics, multiple steady states, and input and output delays [87]. The open-source nature of the CSTR makes it an entry-level benchmark for testing emerging algorithms; using this benchmark, KPCA established the critical role of lag variables [88], and early LSTM-based detectors demonstrated the ability to predict runaway exothermic reactions 30 min in advance.

The third and most recent inflection point occurred in 2015 with the public release of the Cranfield multiphase flow facility benchmark [89], which uniquely couples real experimental data with high-fidelity simulations. Unlike the TEP and CSTR, the Cranfield dataset captures measured pressure and gamma-density signals from physical three-phase flow loops and therefore represents the only openly accessible industrial scale dataset currently available for process fault

Table 6
The comparative analysis of major FDD algorithmic families.

Family	Best-fit conditions	Strengths	Typical weaknesses	Interpretability	Representative interventions
Statistical	stable setpoints; well-calibrated multivariate sensors	Simple, fast; easy alarm integration	limited root cause depth	High (SPE, T^2)	PCA; KPCA
Bayesian	Causal or logic relations available	Probabilistic reasoning; uncertainty quantification	Intensive model tuning; limited scalability	High (causal paths, posteriors)	Bayesian networks
DL	Nonlinear dynamics; rich historians	strongest accuracy on complex dynamics	Opaque; drift sensitive	Low-medium (requires XAI)	CNN; LSTM; AE
Transfer learning	Similar processes across plants; limited target labels	Reduces labeling cost; handles moderate domain shift	Negative transfer if distributions diverge	Medium (domain alignment visuals)	Meta-learning; Transformer-based pipeline
Ensemble	Mixed signals, benchmarks, or multi-modal systems	Stability; robustness; integrates multiple weak learners	Latency of multiple models; monitoring cost	Medium-high (model-agnostic XAI possible)	Causal ensemble explainers

detection. The multiphase system consists of two main units: a two-phase separator and a three-phase separator, with water flow and air flow tracked by adjusting the valve openings. The multiphase flow simulation combined first-principles simulations with sensor data obtained from physical flow loops [90,91], which facilitated its emergence as a significant tool for evaluating process monitoring and fault detection research. In this context, maintaining discussions grounded in publicly available benchmarks not only ensures methodological transparency but also enables the research community to swiftly grasp the challenges of industrial fault diagnosis and initiate reproducible evaluation. As presented in the Appendix (Fig. A1–A3), the detailed process schematics and P&IDs of the three benchmarks are provided to assist potential researchers in replicating or extending these studies.

The eight primary data benchmarks employed in the literature over the past decade are summarized in Table 7, which collectively support most of the quantitative narratives in this review. Among these, the first three entries (S1-S3) are fully open-source simulators that can be obtained at no cost. TEP and CSTR are distributed as MATLAB/Simulink packages via academic repositories, while the Cranfield multiphase flow facility is mirrored on Kaggle with accompanying Python scripts. The sampling frequency in the surveyed datasets extend from a few millihertz to approximately one hertz, about two orders of magnitude. Contrary to high-frequency monitoring of rotating machinery (>1 kHz), continuous chemical processes involve dominant variables such as temperature, concentration, and flow rate that change at relatively slow rates. Sampling once per minute (16.7 mHz) has become the de facto standard configuration, as it provides sufficient bandwidth to capture relevant process dynamics while avoiding unnecessary storage or computational overhead. The benchmarks are also affected by industry bias and fault bias, which require further investigation.

Regarding industry bias, five of the eight benchmarks (S1-S4, S8) originate from the chemical and biochemical industries. The TEP and CSTR simulations provide classic nonlinear feedback behavior, but their application scope remains limited to unit operations and cannot capture complex conditions such as cross unit material cycles or multi-speed disturbances specific to the entire production process. Additionally, the recycling processes involved in water treatment facilities are represented by only two proprietary logs (S6, S7), neither of which are publicly available for download.

As for fault bias, the observable range of fault categories spans from 2 (S5) to 21 (S1), but the number does not equate to coverage. TEP provides 21 fault types and dozens of measurement variables, while these faults cluster around in actuator and sensor failures. Critical initial phenomena for predictive maintenance, such as catalyst deactivation or slow fouling [100], are fairly absent across all datasets.

3.4. Key statistical performance metrics

Statistical performance metrics are reliable tools for converting deviations in massive data into indicators of algorithm effectiveness. Seven types of metrics have gradually become the evaluation basis over the past decade, covering three core engineering concerns: how often faults are caught, how many nuisance alarms are raised, and how quickly and interpretably the system responds. Table 8 below clarifies the specific meaning of these statistical metrics and describes their operational significance, facilitating discussion of the motivations behind their adoption and evaluation of their applicability in industrial process fault diagnosis research.

Accuracy (M1) provides an overall and intuitive measure of classifier success [105], reporting the proportion of correctly labeled samples in the testing set, corresponding to healthy or faulty instances. In industrial settings, however, the fault occurrence rates are assumed to be naturally low, so even a simple model that consistently predicts normal may achieve a deceptively high accuracy. Precision (Pr, M2) and the complementary false positive rate (FPR, M3) rectify this imbalance by quantifying alarm quality: precision expresses the conditional probability that an issued alarm corresponds to a genuine process deviation, whereas FPR tallies the share of routine operating points mistakenly flagged as abnormal. Systems with low Pr (or high FPR) impose a practical burden on control room operators, who must investigate and document each false alarm, potentially eroding confidence in automated monitoring.

Recall (M4) targets the opposite failure mode, the proportion of true faults that are not detected, and is a direct indicator of underlying safety risks. Undetected faults can lead to delayed mitigation measures [106], product non-compliance, and in the worst case, escalate into major hazardous events [3,4,107]. As referenced in [22,60], and [92], Recall is also reported as a synonym for fault detection rate, which are mathematically equivalent and differ only in nomenclature. The F1-score (M5) is defined as the harmonic mean of Pr and Recall, condensing these

Table 7
A summary of data benchmarks for industrial process fault diagnosis.

Study ID	Benchmark	Availability	Data modality†	Fault classes (n)	Sampling frequency	References
S1	Tennessee Eastman Process	Public open	Sim	21	5.56–100 mHz	[73,77,79,92,93]
S2	Continuous stirred tank reactor	Public open	Sim	4–12	16.7 mHz	[29,61,73,87,94]
S3	Cranfield multiphase flow facility	Public open	Hybrid	6 [95]	1 Hz [96]	[30,72,81,83,96]
S4	Fed-batch fermentation penicillin process	Public restricted	Sim	3	13.8 mHz [97]	[39,72,87,98]
S5	Industrial hydrocracking	Proprietary	Real	2	N/A	[20]
S6	Whyalla water reclamation	Proprietary	Real	N/A	16.7 mHz	[99]
S7	Wastewater treatment process	Proprietary	Real	N/A	N/A	[46]
S8	Fluid bed granulation	Proprietary	Real	7	16.7 mHz	[75]

† **Real** = plant/bench-scale sensor traces; **Sim** = first-principles or hybrid simulator output; note **Hybrid** if both are present.

Table 8
An overview of statistical performance metrics for FDD research.

ID	Identifier	Formula	Interpretive meaning	Operational relevance	Key citations
M1	Accuracy (Acc)	$\frac{TP+TN}{TP+TN+FP+FN}\times 100\%$	Overall percentage of samples classified correctly.	Once the faults are very infrequent, this metric can be misleading.	[36,51,52,55,101]
M2	Precision (Pr)	$\frac{TP}{TP+FP}\times 100\%$	Probability that a raised alarm corresponds to a real fault.	High precision suppression of false alarms ensures operator confidence.	[56,61,63,76,102]
M3	False positive rate (FPR)	$\frac{FP}{FP+TN}\times 100\%$	Percentage of samples that are truly normal but are erroneously flagged as fault.	A high FPR produces nuisance alarms in the control room.	[21,22,53,60,103]
M4	Recall	$\frac{TP}{TP+FN}\times 100\%$	Fraction of actual faults that the model captures.	Low recall implies latent hazards.	[31,51,53,54,60]
M5	F1-score (F1)	$2\times\frac{Precision\times Recall}{Precision+Recall}$	Harmonic mean balancing missed detections and false alarms.	Single ranking scalar under fault feature imbalance.	[54,56,61,76,101]
M6	Squared prediction error (SPE)	$\Vert (I-\widehat{P}_i\widehat{P}_i^T)x_i\Vert^2$	The variance of a sample that lies outside the normal operating subspace.	Highly sensitive to subtle, spatially distributed drifts in correlations.	[28,45,48,57,104]
M7	HotellingT^2	$x_i^T\widehat{P}_i\widehat{\wedge}_i^{-1}P_i^Tx_i=x_i^T\widehat{\wedge}_i^{-1}x_i$	Mahalanobis distance of the score vector within the retained component space.	Detects disturbances that remain consistent with the normal correlation processes.	[28,45,48,57,104]

conflicting objectives into a single scalar that penalizes extreme trade-offs, thereby assisting model selection in the absence of an operational cost matrix. Collectively, M1 to M5 characterize the capability of the diagnostic framework to balance plant safety (high M4), economic efficiency (low M3), and operator workload (high M2), providing a multidimensional perspective for robust performance comparisons of industrial process fault diagnosis strategies.

By contrast, the final pair of statistics (M6-M7) measure deviation of the current operating point from the multivariate normal operation state, independent of any class labels. SPE captures the residual variance outside the retained principal component subspace [64] and is therefore good at revealing severe sensor malfunctions, process gain shifts, or unmodeled disturbances. T^2 [108] complements this by calculating the Mahalanobis distance of sample score vectors within the subspace, enabling the detection of subtle, coordinated drifts consistent with underlying pattern changes. Conceptually, T^2 is similar to the multiscale convolutional fault diagnosis method discussed in [94], ensuring that the model captures a broader range of faults across distinct calculation considerations. In general, SPE and T^2 jointly comprise the core of multivariate process fault monitoring and appear together in key citations [28,45,48,57,104]. When any statistic exceeds its control limit [109], the classifier block (M1-M5) can be invoked to identify the most probable fault type, thus achieving a rapid and systematic transition from detection to diagnosis.

4. Challenges and future perspectives

This section consolidates the review's findings on data-driven fault diagnosis and translates them into practical insights. We shift the focus from PRISMA screening to future-oriented exploration, identifying existing applicability gaps and emerging research directions that could reshape real industrial system over the next decade based on the above review results. Three influential themes are emphasized: **LLM-augmented fault diagnostics, benchmark generation via digital twins**, and **explainable fault prognosis and correction**.

4.1. Industrial scenario-dependent and applicability gaps

The comparative analysis in last chapter shows that performance is highly scenario-dependent and shaped by practical constraints. Statistical and Bayesian models still rely on well-calibrated sensors and stationary data distributions [29], making them less effective under multimodal or time-varying conditions typical of chemical and energy plants. The ensemble architectures integrated with DL [70], on the other hand, require extensive labeled datasets and significant computational demand [61], which often exceed what is feasible for on-site or edge deployment. In contrast, transfer learning approaches show promise in bridging cross-plant gaps [69] but still depend heavily on the availability of domain-aligned features and risk negative transfer when domain discrepancies are large.

From an industrial standpoint, each method [71] exhibits a unique balance between interpretability, data demand and operational cost. Classical methods remain attractive for retrofit integration into legacy distributed control systems (DCS) due to their lightweight structure [82] and minimal hardware burden, yet their detection precision deteriorates under strong nonlinearity or sensor noise. Deep and ensemble pipelines deliver superior fault recognition in complex systems [75] but pose challenges in real-time interpretability, model transparency, and hardware compatibility. As industrial monitoring expands toward higher-dimensional data, frameworks must reconcile algorithmic sophistication with operational feasibility.

Evidence remains concentrated on open benchmarks (TEP, CSTR, Cranfield), truly open and plant-grade datasets are scarce [35], so we direct readers to Section 3.3 for a concise accounting of current data availability and bias, as these details will not reiterate here. In light of these challenges, future research must bridge methodological innovation with deployment practicality, which are (i) developing interpretable hybrid frameworks that fuse physical models with LLM for higher transparency; (ii) generating realistic synthetic datasets via digital twins to supplement limited industrial logs; and (iii) embedding adaptive transfer mechanisms to manage domain shifts and data scarcity dynamically. The subsequent Sections 4.2, 4.3 and 4.4 develop these themes with a focus on benchmark generation, explainable and action-linked diagnosis.

4.2. LLM-augmented fault diagnostic

The recent progress of LLM brings a qualitatively new modality to industrial FDD: contextual reasoning based on unstructured process knowledge [110,111]. Fig. 10 sketches the representative LLM-augmented fault diagnostic architecture. The multivariate raw inputs are first converted into low-dimensional embeddings z_t by a lightweight encoder, mirroring the preprocessing adopted for the deep learning pipeline in Section 3.2.4 (Fig. 8). When an abnormal indicator such as SPE is triggered, z_t is concatenated with text segments retrieved from a vector store that indexes historical alarm reports, as demonstrated by Khan et al. [112]. The composite prompt is then injected into the LLM, requiring the model to classify the current system state from a candidate list and provide an explanation in no more than two sentences. The fine-tuning output step can contain scenario metadata,

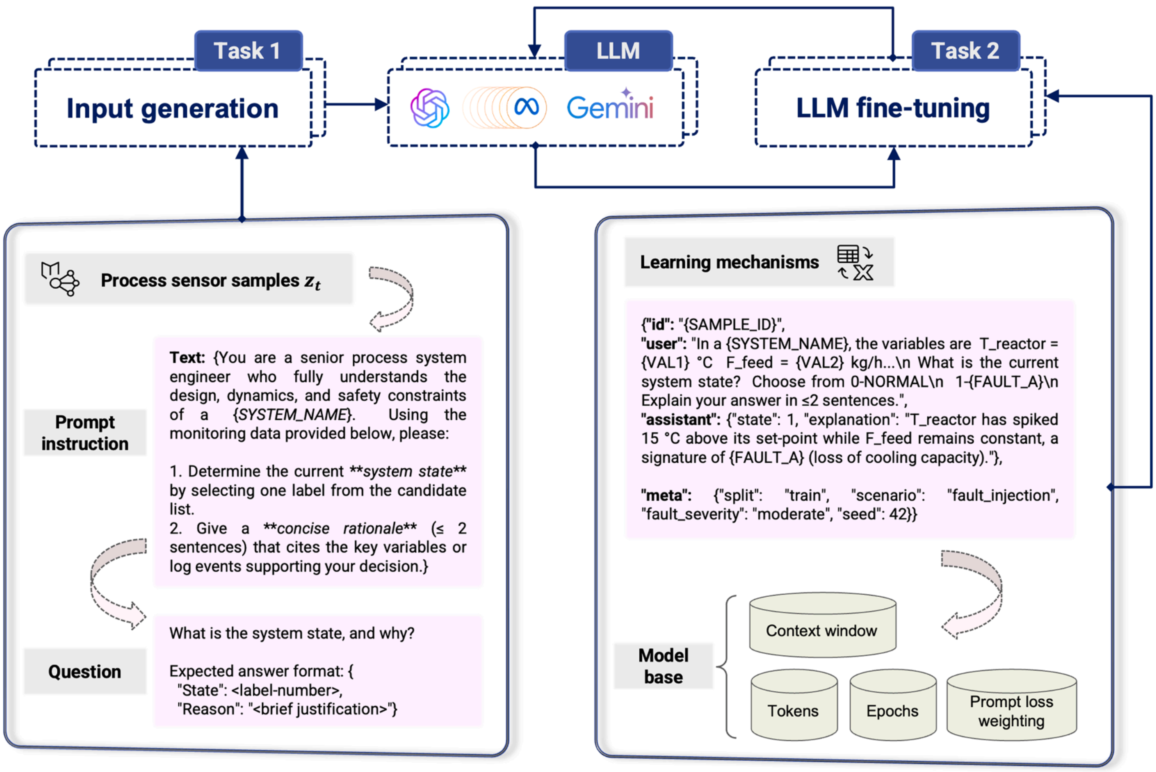

Fig. 10. A multimodal LLM diagnosis diagram: instruction prompts and parameter fine-tuning.

explanatory fragments, and statistical estimates, which has been preliminarily applied to a high-speed train braking system [113], maintaining Acc and F1 above 95 %.

Nevertheless, LLM-augmented fault diagnostic deployment is equally challenging. The pre-trained models may produce illusory content [114] in the form of non-existent valves or sensors, so domain adaptation must be performed on the specific process system corpus. Multimodal alignment between continuous embeddings z_t and discrete text requires contrastive learning or cross-attention mechanisms [115], an area that is still under active exploration. Future research should conduct baseline reliability evaluations of LLM pipelines on publicly available simulations (e.g., TEP, CSTR, and Cranfield), using a combination of F1 and narrative statistical metrics. Moreover, considering the integration of causal graphs reviewed in previous PRISMA screening interventions as prior information could establish a stronger correlation between free-text explanations and the process principles of industrial systems.

4.3. Digital twins in benchmark generation

The comparison element in Section 3.3 reveals that most studies still

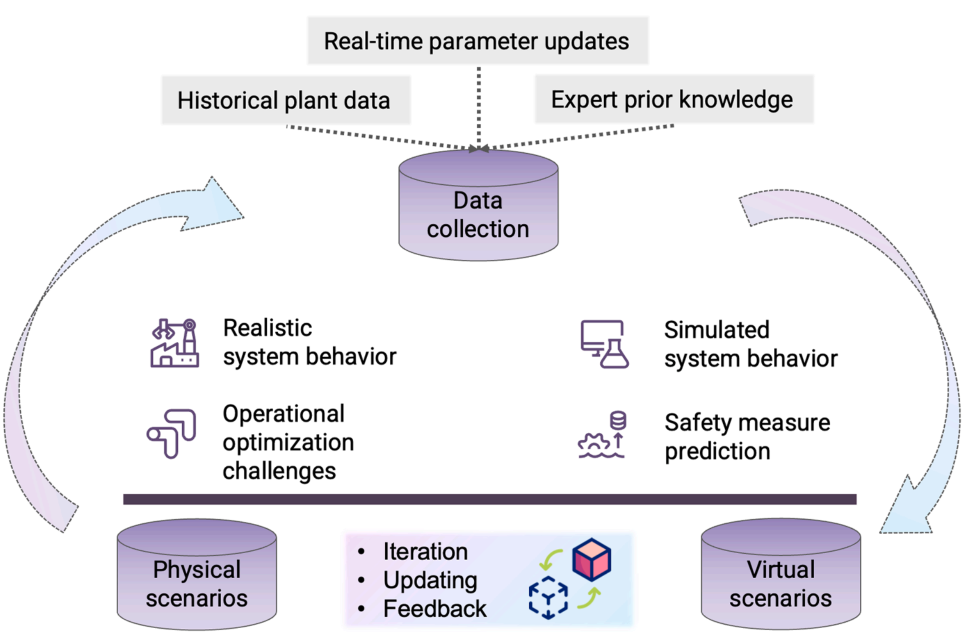

Fig. 11. A closed-loop digital twin framework for industrial process benchmark generation.

rely on three classical simulators, leading to inadequate exploration of unit operations such as reactive distillation and carbon dioxide capture. The DT paradigm offers a potential solution to this limitation, which is defined as the integration of multidisciplinary simulation systems [116]. By leveraging physical modeling and operational history data, it is feasible to construct virtual representations of target industrial processes [117,118]. Fig. 11 illustrates closed-loop bidirectional feedback, where the DT and its physical counterpart continuously exchange information to create benchmark data for FDD research. Scenarios that may involve safety risks in field tests can be mass-generated via DT pattern learning. Crucially, discrepancies between the physical and virtual scenarios can trigger structural optimization of the DT model, thereby enriching the data in an iterative manner through updating and providing feedback on parameters. In a recent attempt, Fang et al. [119] designed a DT model for rotating machinery to accurately describe typical bearing failure modes.

The current challenge is that achieving high model fidelity remains under investigation [120], as unmodeled disturbances or parameter mismatches can propagate unrealistic transients. Computational efficiency is another obstacle, as real-time simulation of large industrial system processes requires high-performance computers with abundant graphics memory to meet data reading demands. In summary, a promising future topic is the application of hybrid digital twins in efficiently generating rare cases in process system historical logs [118,121]. Additionally, coupling DT with transfer learning to generate domain-transformed data is an equally viable research idea. By pre-training DL encoders when target labels are limited, the generalization capability gap mentioned in Section 3.2.5 can be bridged.

4.4. Explainable fault prognosis and correction

The aim of industrial FDD is not merely early alarm generation, but actionable intervention that mitigates economic loss. While Sections 3.2.5 and 3.2.6 document limited explainable progress in explainable artificial intelligence in FDD, our PRISMA evidence reveals probable future developments: explaining fault evolution and prescribing corrective moves that operators can trust [52,122], as shown in the XAI iceberg in Fig. 12. Here, the two load-bearing layers below the decision line deserve special attention. The prescription layer indicates that existing FDD interventions should build the ability to select feasible corrective measures under industrial and safety constraints. The lowest layer illustrates the current general scope of XAI explanations [123], which includes explaining individual data instances and overall modeling algorithms.

The focus of future research is recommended to be on understanding the multilevel linked explanations among fault detection in industrial systems. At the feature level, Shapley additive explanations (SHAP) [79] or local interpretable model-agnostic explanations (LIME) [124,125] can be adopted to quantify the contribution of process variables in prediction degradation. At the prognostic mechanism level, one can investigate how a particular action alters the operational behavior of an industrial system. For instance, the impact of decreasing reflux by 5 % can be analyzed to determine the time required to bring the reactor temperature below the T^2 threshold. In addition, integrating the above findings with the LLM outlined in Section 4.1 can create a reliable dual-channel explanation, thus achieving both fault state prognosis and anomaly root cause tracing.

5. Conclusion

This systematic paper traces the evolution of FDD research in industrial process systems through a rigorous review of 72 articles published since 2015 according to the PRISMA guidelines. By coupling Boolean database queries, inclusion and exclusion strategies, and quality appraisal scales, the proposed framework constructs an evidence base that balances experimental rigor and methodological depth. Quantitative trends show that the number of annual publications reached a turning point in 2021, corresponding to the widespread adoption of DL toolkits. Content analysis provides a structured view of relevant literature in terms of PICO principles: where the research **population** covers multivariate fault diagnosis in continuous and batch chemical process plants; the **intervention** measures include six algorithmic families, ranging from vanilla statistical methods to transfer learning integration; the **comparison** objects investigate actual industrial logs and first-principles simulation benchmarks; and the **outcome** involves standardized metrics including Pr, FPR, and F1, as well as T^2 and SPE thresholds for process monitoring.

Overall, the contributions of this study are summarized as follows:

1) To the best of our knowledge, the present work is a cutting-edge systematic study on fault detection and diagnosis for industrial process systems.
2) A rigorous analysis based on PRISMA reveals emerging trends in industrial fault diagnosis, reinforcing the progress and advances achieved in modeling algorithms.

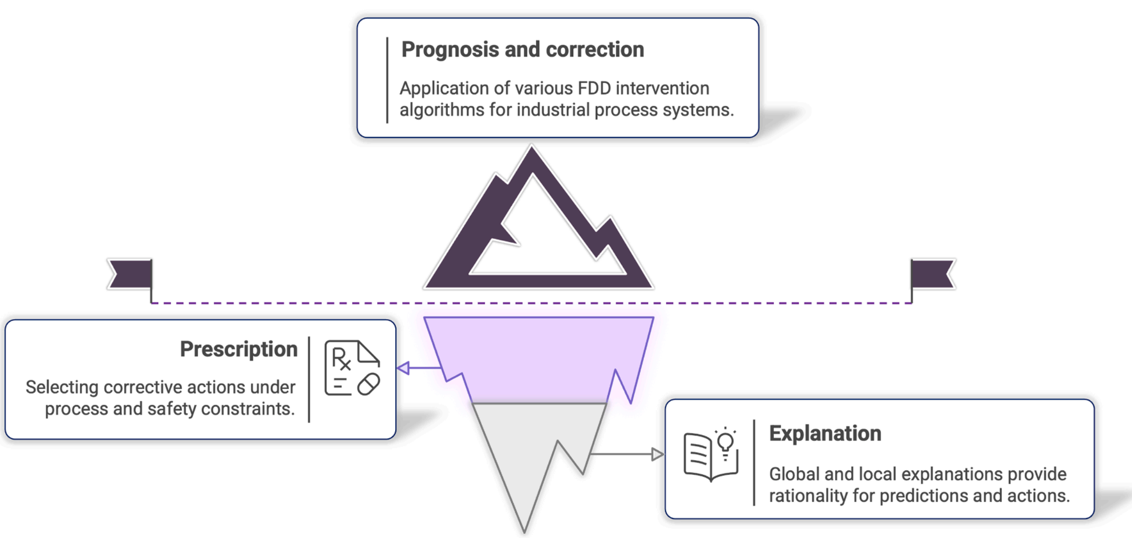

Fig. 12. XAI industrial FDD iceberg supported by prescriptive control and explanatory reasoning.

3) A four-aspect appraisal scale is introduced, adding an additional layer of strict evaluation to the proposed comprehensive review framework.
4) By synthesizing the screened literature discussion, future perspectives and challenges are identified through collective investigation and analysis.

The existing evidence indicates that future FDD research in industrial applications will advance along multiple complementary directions, combining methodological innovation with practical deployment in real process environments. First, hybrid frameworks are expected to replace single models, enabling fault detection systems to perform more reliably under varying process conditions; statistical preprocessing methods are likely to continue delivering the low variance latent spaces, while deep encoders provide the necessary nonlinear capabilities for complex unit operations. Cross-domain adaptation will also become an indispensable technology, encompassing transfer learning pipelines that integrate full lifecycle management and similarity-based diagnosis. Moreover, causal reasoning grounded in LLM and XAI is projected to evolve from emerging concepts into standard evaluation criteria, allowing operators not only to recognize fault conditions but also to interpret their underlying causes and take informed actions to mitigate potential risks. In summary, FDD for industrial process systems is undergoing a pivotal transformation, and future efforts should prioritize the development of algorithmic frameworks that balance reproducibility and interpretability to ensure the sustainable operation of process systems.

CRediT authorship contribution statement

Shuaiyu Zhao: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Haoyu Yang:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis. **Tylee L. Kareck:** Writing – review & editing, Visualization, Validation, Investigation. **Faisal Khan:** Writing – review & editing, Validation, Supervision, Investigation. **Qingsheng Wang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ress.2025.112159.

Data availability

Data will be made available on request.

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